# Melih Tuna İpek

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## SUMMARY

AI/ML Engineer specializing in prompt engineering and applied LLM systems. Currently building the AI core of an AI-powered exam-coaching app as a founding engineer, with prior applied research experience at TÜBİTAK BİLGEM (deep learning, deepfake detection R&D) and backend/LLM exploration at VakıfBank, plus independent projects in agentic AI and computer vision.

## EXPERIENCE

**Part-Time AI Engineer — Amenta** *Jan 2026 – Present*

*Remote · AI-powered coaching app for Turkey's YKS university entrance exam*

- Built the AI core of an AI-powered coaching app for the YKS exam, integrating the OpenAI API via custom Python services and enforcing structured outputs with Pydantic schemas.
- Designed a conversational coaching system with persistent memory: user-shared personal context is extracted and stored in PostgreSQL, enabling personalized, continuous coaching across sessions.
- Architected a 4-phase adaptive study-plan engine driven by exam proximity, student level, and goals; the AI dynamically regenerates weekly plans and triggers phase changes if a student falls behind as the exam nears. Also extended to support Turkey's LGS exam as a planned future feature.
- Built an AI analysis pipeline evaluating student progress, surfacing insights, and triggering proactive notifications, plus a dashboard visualizing progress against the study plan.
- Collaborated in a 5-person founding team since January 2026; used Claude Code for AI-assisted development, GitHub for version control, and Railway for deployment.
- Tech Stack: Python, OpenAI API, Pydantic, PostgreSQL, Railway, GitHub

**Part-Time AI Researcher — TÜBİTAK BİLGEM** *Jul 2024 – Apr 2025*

*Gebze / Kocaeli, Türkiye*

- Contributed to preparation for a deepfake-focused competition, researching and reproducing techniques from academic papers for assigned task items using existing pretrained weights and public datasets.
- Assessed the practical feasibility of each technique under competition conditions — testing whether a competitor could realistically execute it — and containerized working solutions with Docker for reproducible handoff; iterated to alternative methods when one underperformed.
- Occasionally collected supplementary data to test edge cases when existing datasets were insufficient.
- Tech Stack: Python, PyTorch, Docker, Linux

**Software Engineering Intern — VakıfBank** *Jan 2025 – Feb 2025*

*Istanbul, Türkiye*

- Designed and implemented RESTful API endpoints using Java and Spring Framework, validated with Postman.
- Self-initiated exploration of Turkish LLM fine-tuning using internal API access during free time, manually reviewing outputs to assess practical model behavior on Turkish-language tasks.
- Tech Stack: Java, Spring Boot, Postman, Python

## PROJECTS

**Visual Question Answering System**

- Built a VQA system using BLIP to help visually impaired users understand image content through natural-language Q&A.
- Trained and evaluated multiple BLIP configurations on a combined 120K-sample dataset (VizWiz + GQA, ~100K from GQA), benchmarking across question types (color, object, position, text, yes/no) using BLEU, METEOR, and BERTScore; GQA-trained configurations generalized best across metrics.
- Tech Stack: Python, PyTorch, Linux

**Terminal Shopping — Autonomous Product Research Agent**

- Built an autonomous shopping agent that converts natural-language requests (e.g. "iPhone 15 case") into structured search queries using an LLM, then retrieves real-time Amazon results via Apify APIs.
- Designed a review-and-refine loop where the LLM evaluates returned products before presenting them to the user, automatically re-searching if the user rejects the result; succeeded on every product tested.
- Tech Stack: Python, LangChain, Gemini API, Apify API

**mGPT Poet Imitation Fine-Tuning**

- Curated a Turkish poetry dataset of 735 poems across two poets (Necip Fazıl Kısakürek, Pir Sultan Abdal) and fine-tuned the multilingual mGPT model to attempt style imitation.
- Explored parameter-efficient fine-tuning (e.g. LoRA) and prompt-based generation; diagnosed that mGPT's limited Turkish-language capacity, rather than dataset size, was the primary bottleneck to coherent style transfer.

- Tech Stack: Python, PyTorch, Hugging Face Transformers, LLMs, NLP

**Autonomous Driving Simulation**

- Collected training data by manually driving a simulated car (steering left, right, or straight) while recording camera frames paired with each input.
- Trained a CNN based on NVIDIA's end-to-end self-driving architecture to predict the correct steering action (left / right / straight) directly from camera frames, successfully replicating human-like steering decisions in the simulation.
- Tech Stack: Python, TensorFlow

**Binary Image Reconstruction with Genetic Algorithm**

- Implemented a genetic algorithm approximating binary images using line segments across 360 circle points, converging to 95% pixel similarity.
- Tech Stack: Python, OpenCV, Pillow

## TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, Dart
AI: LLM, Prompt Engineering, Computer Vision, VLM, LangChain, Agent Orchestration
ML / Data Science: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, pandas, NumPy, Matplotlib
APIs & Data: OpenAI API, Pydantic, PostgreSQL
DevOps & Tools: Git, Docker, Linux, Postman, Railway
Frameworks: Spring Boot, Flutter

## EDUCATION

**Yıldız Technical University — B.Sc. Computer Engineering** *2020 – 2025*
*İstanbul, Türkiye · GPA: 3.23 / 4*

**Kırıkkale Science High School** *2016 – 2020*
*Kırıkkale, Türkiye · GPA: 94 / 100*

## CERTIFICATIONS

Deep Learning Specialization — DeepLearning.AI (Coursera), Jan 2024
Machine Learning Specialization — DeepLearning.AI (Coursera), Aug 2023

## LEADERSHIP & ADDITIONAL

- SKY LAB — AI Education Team Lead / Member: led and contributed to AI-focused learning and project activities within a student community.
- Solved 90+ LeetCode problems in C++, strengthening algorithmic and data-structure fundamentals.

## LANGUAGES

- English — C1 (YDS: 82.5, IELTS: 7.0/9.0)
- Turkish — Native